

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458293.8096 +/- 0.003 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.15 +/- 0.0097
Transit depth (Rp/Rs)^2	2.25 +/- 0.29 [%]
Orbital Inclination (inc)	88.69 +/- 1.02
Airmass coefficient 1 (a1)	1.35 +/- 0.019
Airmass coefficient 2 (a2)	-0.2462 +/- 0.0059
Scatter in the residuals of the lightcurve fit is	1.36 %
Best Comparison Star	None
Optimal Aperture	4.28
Optimal Annulus	6.41
Transit Duration (day)	0.0971 +/- 0.0024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.15 ± 0.0097 vs 0.1326 (deviation 1.47σ)
Duration fit vs published	0.0971 d vs 0.0737 d (deviation 7.97σ)
Mid-time O–C	3.6 min
Depth SNR	12.7
Residual scatter	1.36 %

Light curve

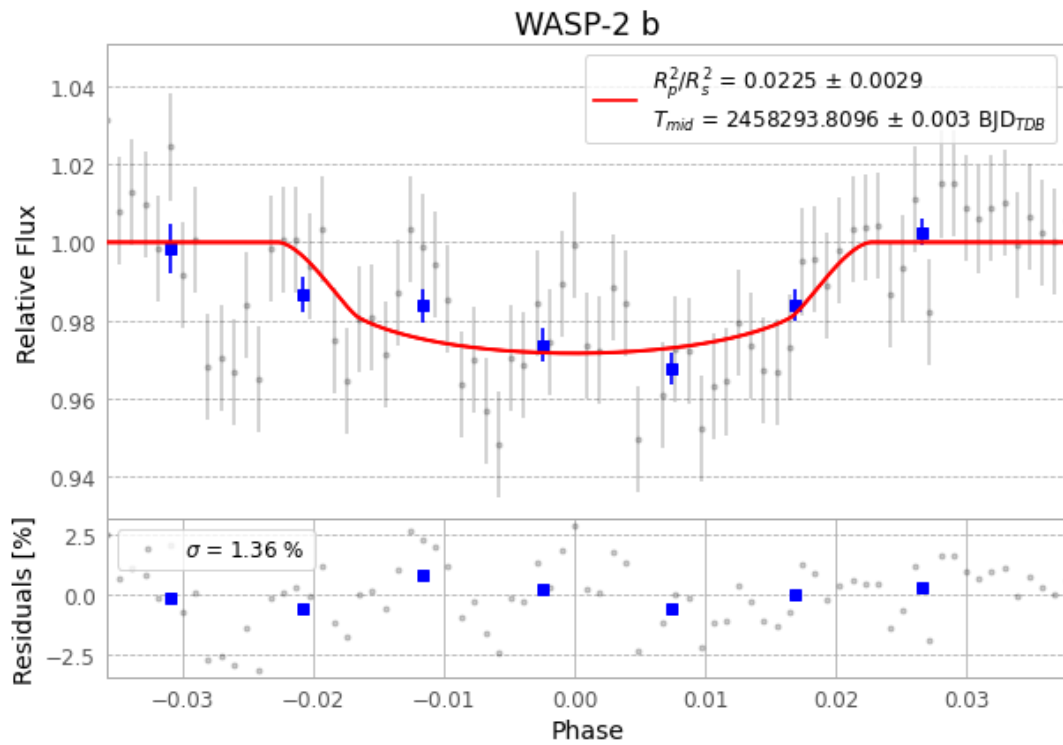


Figure 1 — FinalLightCurve_WASP-2 b_2018-06-23.png (SHA-256 aa91ac54a38f21c6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [341, 145], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e8e3f9fd2bef708d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 692f538

Data provenance

77 FITS frames (51 MB), WASP-2180624052712.FITS → WASP-2180624091512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-180624.log (860e22965341...), FinalLightCurve_WASP-2 b_2018-06-23.png (aa91ac54a38f...), FinalLightCurve_WASP-2 b_2018-06-23.csv (a9830c68c045...)

Compute resources

Wall time 135.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 16:35Z.